

科目 計算機概論 類組別 A4

單選題，共 20 題，每題 5 分（答錯不倒扣）

1. Convert the decimal number -6.5 into IEEE-754 single-precision floating-point format. What are its sign bit, exponent field, and the first 4 bits of the fraction field, respectively? Assume that the fraction field does not include the hidden leading 1.

- (A) 0, 10000001, 1010
(B) 1, 10000010, 1010
(C) 1, 10000001, 1010
(D) 1, 10000001, 1101
(E) 1, 01111110, 1010

$$6.5 = 110.1 = 1.101 \times 2^3$$

$$2^3 - 1 = 2 + 1/2 + 1/4 = 1.75 = 10000001$$

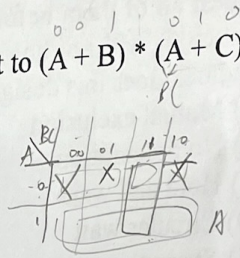
2. Which of the following Boolean expressions is equivalent to $(A + B) * (A + C)$? Here, + denotes OR, and * denotes AND.

- (A) $A + B + C$
(B) $A + B * C$
(C) $A * B + C$
(D) $A * B * C$
(E) $A * (B + C)$

$$[A * (A + C)] + [B * (A + C)]$$

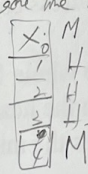
$$= A * A + A * C + B * A + B * C$$

$$= A * (1 + C + B) + B * C = A + B * C$$



3. A byte-addressable computer stores an integer array A, where each integer occupies 4 bytes. The cache line size is 16 bytes, and A[0] starts exactly at the beginning of a cache line. The program accesses the elements in the following order: A[0], A[1], A[2], A[3], A[4]. Assume the cache is initially empty, the cache is large enough, and no cache line is evicted during these accesses. Which hit/miss pattern is expected? Use M for a miss and H for a hit.

- (A) M, M, M, M, M
(B) M, H, H, M, H
(C) H, H, H, H, H
(D) M, H, M, H, M
(E) M, H, H, H, M



$$16 \text{ GB} = \frac{2^{34} \text{ B}}{1 \text{ B}} = 2^{34}$$

4. A computer system is byte-addressable and has a maximum addressable memory space of 16 GB. If each byte requires a distinct address, how many address bits are needed at a minimum?

- (A) 30 bits
(B) 32 bits
(C) 33 bits
(D) 34 bits
(E) 36 bits

注意:背面有試題